

Active Inference ~ Parr, Pezzulo, Friston ~ Chapter 2

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on to chapter two the low road to octave inference so all chapters begin with a short quotation and the quotation here reads my thinking is first and last and always for the sake of my doing William James so even before reading the chapter these quotations are often great places to jump off and have a discussion so Ali please begin and go as long as you want on the low road to active inference okay so uh yes chapter two uh as the title suggests it's the construction of active inference Theory from the Viewpoint of of the low road active inference So It Begins by uh providing this notion of perception as inference because uh more often than not we usually think of perception as something uh just in a computational sense of the sense is the processing of the of the inputs or rather it's the raw processing of the inputs but here it shows that it can also be described as or even more accurately described as a kind of inference so it's not just a simple and raw processing of the inputs as my a computational model uh or a computer analogy might suggest it's more um like uh something that we predict and then um then we compare our inputs or the inputs we get from the stimuli uh with our prediction and then try to somehow minimize that prediction error so I believe that this notion of perception is inference is uh the most Central notion of all the related theories of predictive coding predictive processing Bayesian brain hypothesis so up to now active inference uh is not very different from all the other theories it's basically a subset or a variant of those theories but then it progresses uh into uh distinguishing uh distinguishing the active active inference from all the other theories and how uh it stands apart from the other ones uh so uh in section 2.1 uh we began with some of the basics of probability Theory namely the base theorem uh in box 2.1 uh and uh some simple examples uh about how base theorem can be applied uh and again a very important uh page or important part of this section will be page 20 which describes the concept of surprisal and specifically the statistical surprisal and how it differs from the phenomenological surprise or psychological surprise uh so uh and that's how we can formulate

the statistical surprise yes yes just just to catch up here this example that's going to come back again and again is a person who's guessing the object that's in their hand and it could be a frog or an apple and the object is going to jump or not and so that's used as a way to talk about the Bayesian updating that forms the kernel of a variety of Bayesian brain like models including active inference and then um there are the presentation of the exact base so in simple cases you can compute exactly what you want with Bayes theorem however it's a lot more effective on large data sets to use certain approximations and heuristics that we're going to be discussing um Ali has highlighted that there are two concepts that are very closely related to each other regarding surprise there's Surprise by itself and then there's Bayesian surprise so please pick up there what are surprise and Bayesian surprise and why do they matter here yes so um well surprise I mean the regular surprise as we're all familiar with uh is something um more related to our sense of surprise or psychological sense of surprise of uh observing some unexpected phenomena or unexpected Behavior but based in surprise or statistical surprise uh is the of course closely related with the psychological sense of surprise but a bit more rigorous in which it's a way to compare to um probabilistic information using a callback Library Divergence and somehow getting the I mean how unexpected that information emerging from callback Divergence uh and uh they call it the surprisal so uh surprisal is uh you know in other words is a way to formulating those unexpected uh probabilistic information and it doesn't necessarily uh align or maps onto perfectly are psychological sense of surprisal but as I said it's closely related with that I know I was just gonna say that um I was just gonna say that by surprise um almost in the rest all the um other sections and chapters of the book uh we basically mean this Bayesian surprise uh sense of the world and in order to distinguish that with the regular uh sense of surprise uh sometimes in the literature surprisal is used to refer to Bayesian surprise yes so they're both measured in information theoretic units this is all happening in information geometric spaces the first concepts of surprise is applied to a given single observations how surprising is that one observation and so in that sense it's a lot like the z-score of a data point coming in with respect to a statistical distribution so it's like you have a height distribution in a classroom and you measure one person and then you can say what is the z-score of that measurement was it right at the center of the distribution with the z-score of zero or with a two standard deviations higher with a z-score of two so it's kind of like that and that's why there's a discussion of probability distributions and their support which is the X values for which they're defined and surprise function with the fancy I and that's going to be a function that helps you compute how surprising each observation is given a parameter as of

that distribution whereas this Bayesian surprise is more related to learning it has to do with how much updating happens between the prior and the posterior and that's before and after the observation so one could imagine a surprising observation surprise concept one that either does or doesn't update the prior into a very different posterior so as Ali mentioned they are not exactly the same but it's going to be important to understand how they're different and that's worked out again in the case of the Apple jumping box 2.2 continues with a discussion of expectations now expectation in everyday parlance might be specifically referring to something in the future like I expect it to rain tomorrow and in statistics when we talk about the expectation we're talking about the weighted average or the center of gravity of a distribution and that can be in both a discrete distribution at which point you have a weighted sum or a continuous distribution in which case it's an integral so you can have an expectation for the humidity tomorrow and so that might refer to the center of gravity at $A \cdot T$ equals plus one but just taking a loan expectation means center of gravity of a statistical distribution not anticipation section 2.3 is going to describe how some of this how this low road that we're on is going to connect to biological inference there's uh more discussion of the generative model and the generative process and a little bit of a hint that the generative model captures aspects of the generative process which is where we see a lot of the classical cybernetic theorems and Concepts like requisite diversity good regulator theorem and so on however the generative model does not have to be exactly isomorphic with the generative process for example degenerative process the temperature in the room might be a continuous variable but then the generative model might be discrete only modeling integer based temperatures or might be categorical like too hot just right and too cold so there's a lot of articulations that can be done because of how flexible and interoperable generative models are with each other figure 2.2 is going to expand upon that chapter 1 representation of the cybernetic action perception Loop and we're going to see this more in terms of a generative model and generative process articulation and there are incoming speaking from the perspective of the generative model the agent they're incoming and outgoing dependencies in this graph and this is a little bit like a schema more so than a formal graph but also it is like a Bayesian graph where nodes are variables and edges are causal relationships and so we have the internal states of the model the external states of the generative process and then the blanket states that make internal and external States conditionally independent and again speaking from the perspective of the agent although there is a symmetry we can talk about the incoming sensory signals happening from the observations handed from the process passed on to the internal states of the generative

model and then the outgoing actions that are selected that then can influence the hidden state of the world for example going and turning on the heater to increase the temperature in the room and so this action perception Loop or particular partition is going to get explored in a lot more detail in the coming sections previous section 2.3 was on perception as inference and now action as inference is going to be discussed and that is where they say the discussion to this point is common to all Bayesian brain theories we now introduce the simple but fundamental Advance offered by octave inference which is the extension of this inferential perspective to consideration of action as inference perception action cooperate to realize a single objective section 2.5 is about minimizing the discrepancy between the model and the world we already described that there's two ways for this to happen change your mind and change the world that's how the discrepancy can be reduced or managed we see a variant of the action perception Loop where the agent is making perceptions uh predictive models of the world that through perception are being juxtaposed with observations handed from the generative process the world and a discrepancy is realized some non-zero discrepancy and then here are those two paths to minimize free energy change beliefs by perception and learning or change the world through action section 2.6 is going to discuss how the exact Bayesian approach described earlier is absolutely spot on if you have infinite Computing resources however we're often interested in Rapid or large data sets where we want to be able to get approximate Bayesian computation or probably approximately correct computation in a vastly accelerated fashion and so that is going to be approached using what's called variational Bayesian inference that's unpacked in chapter four but it suffices to say that variational Bayesian inference implies substituting two intractable quantities the posterior probability and log model evidence with two quantities that approximate them but can be computed efficiently the approximate posterior q and a variational free energy so it transforms an intractable estimation problem into a highly tractable optimization problem pick up from there Ali uh yes but sorry I just wanted to point out out a couple of things especially about section 2.3 because I believe it is um one of the crucial sections of this chapter and in fact the whole book because uh it provides some of the justifications of using uh Bayesian inference uh as opposed to as some other uh mathematical techniques such as maximum likelihood estimation but more important than that I think it's this concept of optimality which comes into play in almost everywhere in the literature and of course in this book so there are actually two Notions of optimality uh which is not discussed in detail here namely Bayesian optimality and Jane's optimality but uh some in some of the recent papers and Bayesian mechanics uh Dalton's

active at development and others have shown that those two concepts of optimality are actually congruent with each other so that's one of the reasons that the Duality between free energy principle and constrained maximum entropy principle I can I mean it's one of the justifications for uh providing that dual formalism between those two uh so uh but another point I wanted to mention here is because I've seen that using uh the word hidden State uh can be a bit confusing for some people because uh when we observe uh something as an observation obviously it is not quote-unquote hidden right so what a hidden State here refers to is actually the hidden cause of that observation so it's not that the observation itself is hidden from the observational or it's unobserved so that might be a bit uh confusing if if we don't take into consideration the exact meaning of the Hidden state or latent State here and in the rest of the literature so continuing from section 2.6 uh here we see uh one of the two Central equations of active inference which is equation 2.5 for variational free energy so it's as I said I mean understanding this equation and how each line of its formulation represents in terms of the uh I mean trade-off between energy entropy or complexity inaccuracy or Divergence and evidence is key to understanding almost everything in the rest of the book and in many other literature on active inference so this is the perceptual part of active inference so variation free energy is a parameter that uh uh is a notion that parameterizes the surprisal of our perceptual information we have about the external States and then we'll see in the next section uh the related and almost symmetrical formulation to variation of free energy namely expected free energy which is basically the action part of the active inference so we can see how those two uh can somehow uh be seen as a kind of unified formalism but described in the alternate Expressions so but another thing about equation 2.5 is uh it it may be a bit um I don't know daunting to see all the relations between those three lines of equation and how we can get from one to the to the other so there are some supplementary materials that we have developed in the past weeks which I think can help in clarifying how the derivative nations of these three lines of equation be done so I hope they would be clarifying in I mean and helps to understand how those three lines relate to each other uh so but the key Point here is uh to understand the variation of free energy is not something uh absolute but it's just an upper Bound for the minimization so as Daniel just mentioned it's uh untractable to have an absolute amount for the surprisal to be minimized so we need to have an upper Bound in order to make that more tractable because uh by Jane's inequality as we'll saw in um we chapter four I think we can see that how the upper bound of the surprisal uh necessarily provides a condition for minimizing the precise or the exact um free the exact free energy and that's the key Insight of

equation 2.5 or the notion of variation of free energy which is to provide this upper bound instead of the exact amount of surprises to be minimized yeah what I'll what I'll add there is if you knew exactly how well you should be surprised by a given data point why then you would have had the optimal model however that is not tractable and so by making a quantity that's always higher or an upper bound the variational free energy F which is a function of broadly Q our beliefs or variational beliefs which are built in a way that makes them very compositional very optimizable very interpretable and data and we can reduce the Divergence here are the KL divergences with a double line between Q our beliefs and P the kind of actuality of it and if we can reduce this Divergence in other words minimize the free energy then we will come closer and closer to the true surprise function and do that in an attractable incrementally optimizable way so equation 2.5 is going to be the variational free energy different ways that it can be represented as it takes in data and beliefs about the world Q and then this is going to be expanded into the future to include action with G the expected free energy now there's a lot more that we can say about this there's a lot of technicalities to go into but broadly notice that g the expected free energy is a functional of policy π_i because it's only being evaluated to select amongst different action outcomes π_i and another important difference is that it's going to be describing sensory outcomes that haven't yet happened a sort of what would I perceive if I did a or what if I did B and it's that kind of comparison that allows the expected free energy functional here to be used in action selection or policy selection as inference planning as inference that section is expanded upon and in figure 2.6 we see a very nice representation of the expected free energy equation and then how when certain aspects of this equation or situation are zeroed out we get certain other familiar cases for example where there's no epistemic value there's no information to learn then you get ruthless expected utility Theory and conversely where there is no pragmatic value to extract so all outcomes are equally valid for or preferable then you get things like Infomax principle and optimal Bayesian design and then everything in between is the space that we're interested in and so this figure 2.6 shows that the expected free energy functional can be seen as like a generalization of a lot of other settings related to perception and action and planning amidst uncertainty and section 2.9 closes the low road they took us all the way to active inference from Bayes theorem through the generative model onto active inference and clarifies these two Notions of variational free energy that's the real-time perceptual unfolding evidence lower bound on surprisal tractable optimizable and so on and f and g the expected free energy which is able to do planning as inference or policy selection as inference expected free energy is fundamentally prospective

and that enables counterfactual cognition section 2.10 summarizes active inference is the theory of how living artifacts underwrite their existence by minimizing surprise or attractable proxy to surprise variational free energy via perception and action and they motivate that from a first principles base theorem starting place any closing thoughts on chapter 2 Ali I would just humbly suggest for the people I want to go through this chapter to try try their best to really understand uh specifically what equation 2.5 and 2.6 represents and how uh I mean how to use those equations to describe different situations with some missing elements as well because I believe those sections and particularly those equations are absolutely essential to for understanding uh everything active in prints related both in this the rest of this book and uh in the rest of the and in almost all thanks in just the last thought I'll give on chapter two is this is exactly the work that we do in the active inference textbook group we really welcome all backgrounds every single question and uncertainty you have is beautiful we have a lot of resources that Ali and others make to make the math approachable and rigorous and natural language descriptions of the equations and so on so yes it's really important to understand the equations because after all that's like the skeleton that gives meaning to our usage and fluency of the active inference ontology which Ali and I are speaking right now we're not just saying surprise is related to this because we felt it that way it there is an underpinning and it is a really interesting life's work to explore it but we're finding ways to communicate it and learn and teach it better and better every time so that's chapter two that's the low road